

Sprint 1

Overview

Write a Python program that will interpret pseudo-English statements describing an amount of money in coins and convert them into a dollar amount

Functional Requirements

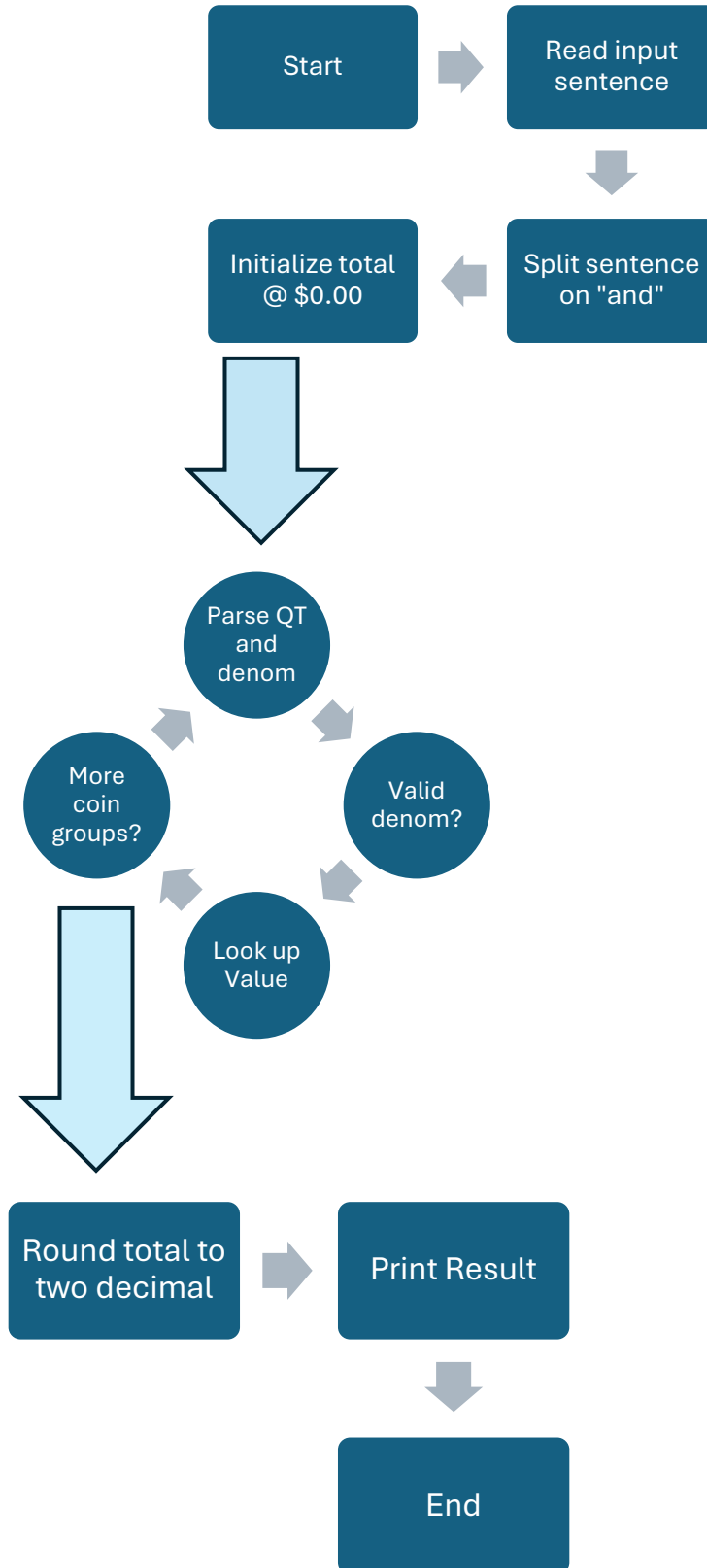
Data Input:

- The system should accept input sentences describing coin quantities and denominations in pseudo-English.
- Each input should consist of one or more coin groups, where each group contains a numeric quantity and a coin denomination, such as pennies, nickels, dimes, etc.
- Multiple coin groups can be joined with the word “and”.
- The system should recognize both singular and plural forms of each denomination.
- The system should convert each coin group to its dollar value using the following rates
 - Pennies = \$.01
 - Nickels = \$.05
 - Dimes = \$.10
 - Quarters = \$.25
- The system should be able to process input as a string.
- The system should be able to handle any number of coin groupings in a single input.
- The system should be able to interpret these without any standard English punctuation being used.

Data Output:

- The system should total all coin group values and output the total as a dollar amount rounded to two decimal places.
- The system should print results to the console.

Program Flow



Test Case Results

- 1 penny and 2 nickels -> 0.11
 - **Pass/Fail**
- 4 dimes and 7 quarters -> 2.15
 - **Pass/Fail**
- 1 quarter and 3 pennies -> 0.28
 - **Pass/Fail**
- 21 pennies and 17 dimes and 52 quarters -> 14.91
 - **Pass/Fail**
- 95 dimes and 73 quarters and 22 nickels and 36 pennies -> 29.21
 - **Pass/Fail**
- 1 nickel and 17 quarters -> 4.30
 - **Pass/Fail**
- 21 nickels and 15 pennies -> 1.20
 - **Pass/Fail**
- 1 dime and 1 nickel and 1 penny and 1 quarter -> 0.41
 - **Pass/Fail**

```
C:\WINDOWS\system32\cmd. x + v
=====
WELCOME TO THE COIN CONVERTER PROGRAM
This program converts a sentence describing coin quantities into a total value in dollars.
Type 'quit' to exit the program.
=====

Enter your coin sentence: 1 penny and 2 nickels
Result: 0.11

Enter your coin sentence: 4 dimes and 7 quarters
Warning: Unknown denomination 'quarters'skipping
Result: 0.40

Enter your coin sentence: 1 quarter and 3 pennies
Result: 0.28

Enter your coin sentence: 21 pennies and 17 dimes and 52 quarters
Warning: Unknown denomination 'quarters'skipping
Result: 1.91

Enter your coin sentence: 95 dimes and 73 quarters and 22 nickels and 36 pennies
Warning: Unknown denomination 'quarters'skipping
Result: 10.96

Enter your coin sentence: 1 nickel and 17 quarters
Warning: Unknown denomination 'quarters'skipping
Result: 0.05

Enter your coin sentence:
Please enter a sentence.

Enter your coin sentence: 21 nickels and 15 pennies
Result: 1.20

Enter your coin sentence: 1 dime and 1 nickel and 1 penny and 1 quarter
Result: 0.41

Enter your coin sentence: |
```

Sprint 2

Overview

Write a Python program that will interpret pseudo-English statements describing an amount of money in coins and convert them into a dollar amount

Functional Requirements

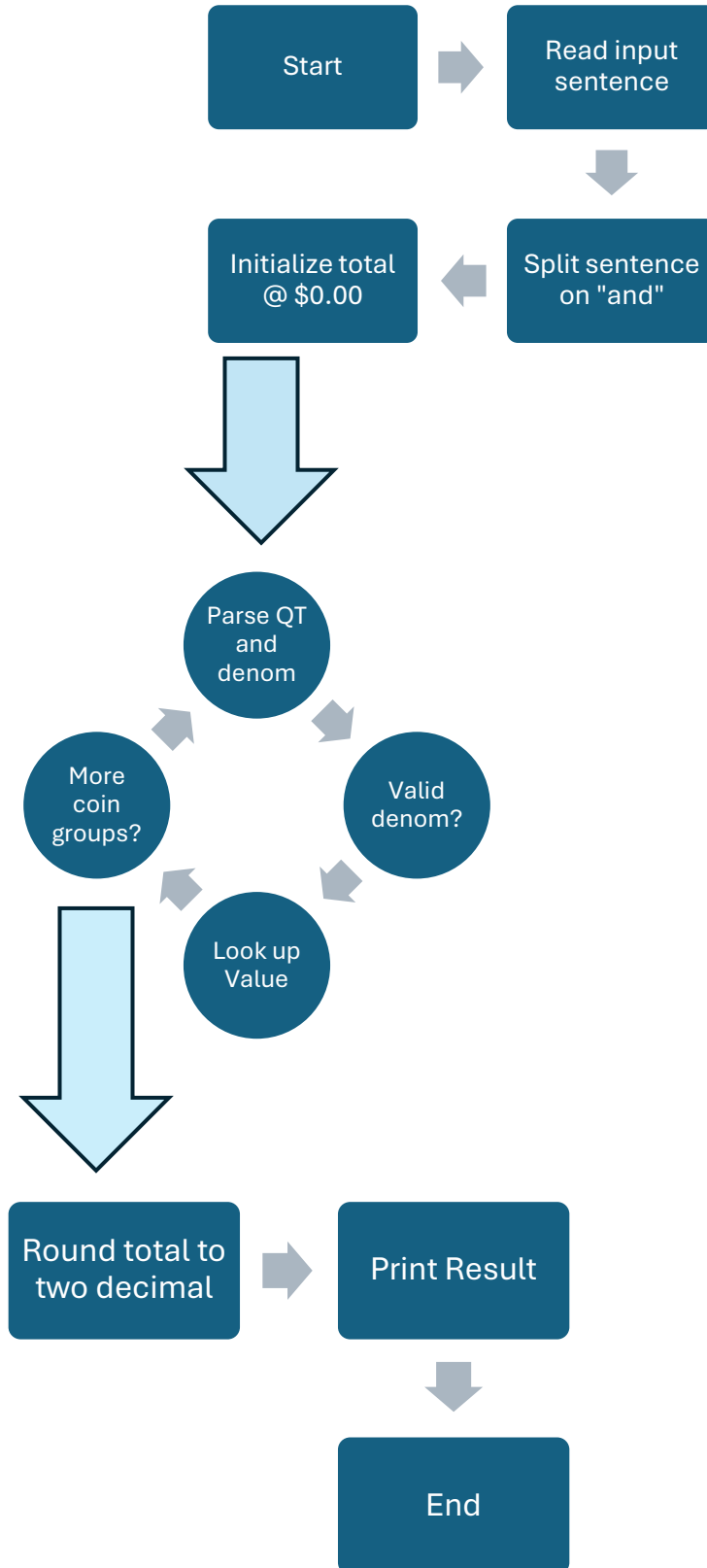
Data Input:

- The system should accept input sentences describing coin quantities and denominations in pseudo-English.
- Each input should consist of one or more coin groups, where each group contains a numeric quantity and a coin denomination, such as pennies, nickels, dimes, etc.
- Multiple coin groups can be joined with the word “and”.
- The system should recognize both singular and plural forms of each denomination.
- The system should convert each coin group to its dollar value using the following rates
 - Pennies = \$.01
 - Nickels = \$.05
 - Dimes = \$.10
 - Quarters = \$.25
- The system should be able to process input as a string.
- The system should be able to handle any number of coin groupings in a single input.
- The system should be able to interpret these without any standard English punctuation being used.

Data Output:

- The system should total all coin group values and output the total as a dollar amount rounded to two decimal places.
- The system should print results to the console.

Program Flow



Summary of Bug

I had quarter listed twice in the original code, which resulted in the unknown denomination message appearing on several of the test cases. Fixing and running back through the same test cases shows resolved.

```
C:\WINDOWS\system32\cmd. X + -
=====
WELCOME TO THE COIN CONVERTER PROGRAM
This program converts a sentence describing coin quantities into a total value in dollars.
Type 'quit' to exit the program.
=====

Enter your coin sentence: 1 penny and 2 nickels
Result: 0.11

Enter your coin sentence: 4 dimes and 7 quarters
Result: 2.15

Enter your coin sentence: * 1 quarter and 3 pennies
Warning: Skipping unrecognized group '* 1 quarter'
Result: 0.03

Enter your coin sentence: 1 quarter and 3 pennies
Result: 0.28

Enter your coin sentence: 21 pennies and 17 dimes and 52 quarters
Result: 14.91

Enter your coin sentence: 95 dimes and 73 quarters and 22 nickels and 36 pennies
Result: 29.21

Enter your coin sentence: 1 nickel and 17 quarters
Result: 4.30

Enter your coin sentence: 21 nickels and 15 pennies
Result: 1.20

Enter your coin sentence: 1 dime and 1 nickel and 1 penny and 1 quarter
Result: 0.41

Enter your coin sentence: exit
Warning: Skipping unrecognized group 'exit'
Result: 0.00

Enter your coin sentence: quit
Goodbye!
Press any key to continue . . . |
```